

Assignment 4

Residuals, LayerNorm & Gradient Flow in Depth

Strict Protocol Edition · Towards Research-Level Competence

Assignment 4

■ GLOBAL RULES

- Form a hypothesis BEFORE every experiment.
- Explain your reasoning before asking questions.
- After using external help, re-implement from memory.
- Every experiment must include: Hypothesis → Result → Interpretation.

Problem Statement

Stack 2-4 attention layers and systematically study how gradient flow changes with depth. Add residual connections and LayerNorm one at a time, measuring the effect of each component independently before combining them.

- Task: multi-step reasoning -- given a sequence, output the max of the bottom half and min of the top half
- Architecture: 2-4 stacked attention layers, embedding dim ≤ 64
- Four ablation runs: (1) no residuals/no LN, (2) residuals only, (3) LN only, (4) both
- Measure gradient norms at each layer for all four configurations
- Identify the depth at which vanishing/exploding gradients appear without residuals

Implementation Requirements

1. Implement a stack of N attention layers with configurable residuals and LayerNorm
2. LayerNorm must be implemented manually (compute mean, variance, scale, shift yourself)
3. Train all four ablation configurations from the same random seed
4. Log per-layer gradient norms at every 100 steps
5. Plot the ratio of gradient norm at layer 1 vs. layer N over training

Instrumentation Requirements

1. Plot gradient norms per layer across training for all four configurations
2. Track output distribution at each layer -- does it collapse or explode without LN?
3. Measure final accuracy and convergence speed for each ablation
4. Log the LayerNorm learned scale (gamma) and shift (beta) over training

Failure Analysis (Required)

You MUST document failures. Describe what failed, why you think it failed, test at least two fixes, and evaluate which fix worked and why.

1. At what depth does the no-residual model fail? Is it sudden or gradual?
2. Does LayerNorm alone (without residuals) stabilize training? Why or why not?
3. Do residuals alone (without LayerNorm) prevent vanishing gradients?
4. What happens to gradient norms in the full model (residuals + LN) -- are they truly stable?

Mathematical Integration

1. Derive why residual connections mitigate vanishing gradients (show the gradient path)
2. Compute the effective depth of the network -- how many layers actually receive gradient?
3. Analyze the Jacobian of each layer with and without residuals
4. Study the loss surface curvature: is it smoother with LN? Use finite-difference approximation.

Research Questions to Answer

- Is there a minimum depth needed to solve the multi-step task? Why?
- Does adding depth always help, or does it plateau? Where is the inflection point?
- What does a stable gradient norm ratio look like across layers? What is pathological?
- Can you predict from the gradient norms alone whether a run will converge?

Phase Progression

Phase	Focus
Phase 1	Build the core architecture
Phase 2	Compare with baseline / non-attention models
Phase 3	Study optimization dynamics
Phase 4	Modify loss functions and evaluate

Phase 5	Alter architecture systematically
Phase 6	Design novel training techniques

Document everything like a research paper. Your goal: independently analyze neural architectures, diagnose training failures, and propose novel improvements grounded in theory and empirical evidence.